



$$\sigma_i = \pm 1/2$$

$$L = \gamma \mu_B B \sum_i \sigma_i = \frac{1}{2} \gamma \mu_B B \sum_i \sigma_i$$

$\uparrow$ $\uparrow$
 μ_B γ

MEAN FIELD THEORY

STEP 2

MATHS
201

$$\langle \sigma_i \rangle = \langle \sigma_j \rangle = \sigma$$

$$\sigma = \frac{1}{2} \tanh\left(-\frac{\beta J}{2} \sigma + \frac{\beta g \mu_B B}{2}\right)$$

$$m = -J \sigma$$

B=0

$$\sigma = \frac{1}{2} \tanh\left(\beta \frac{J \sigma}{2}\right)$$

$$y = \sigma$$

$$y = \frac{1}{2} \tanh\left(\beta \frac{J y}{2}\right)$$

